Nadir ZOUAOUI

Electrical & Instrumentation Engineer - Oil & Gas, Hazardous Areas, Commissioning

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Class B driver's license | Personal vehicle | National service: exempt

PROFESSIONAL SUMMARY

State engineer in electrical engineering with four years of experience on Algerian Oil & Gas, industrial and renewable projects. Sonatrach SCADA pipeline projects, ATEX Ex "d" and Ex "p" designs, commissioning of 27 chemical injection skids and their solar power supplies, and loop checks on the ALRAR gas complex. Available for site work and desert rotation.

CORE SKILLS

Oil & Gas facilities | Hazardous Areas (ATEX / IEC 60079) | Commissioning and loop checks | MCC & Switchgear | SCADA and industrial automation | Electrical design and schematics | Off-grid and grid-tied PV power supplies

TECHNICAL TOOLS

EPLAN Electric P8 + Pro Panel | AutoCAD Electrical | Caneco BT | Archelios PV simulation | TIA Portal, STEP 7 | Siemens S7-1200 | Microsoft Excel | Blender 3D

CERTIFICATIONS

Electrical Equipment in Hazardous Areas (EEHA) - ROAR Training Solutions (Australian RTO)

2026 | Online | Australian national competency units (AS/NZS 60079, aligned with IEC 60079)

- Classified hazardous areas, explosion-protection requirements and planning of Ex installations (UEEHA0004, UEEHA0022, UEEHA0029).
- Installation of explosion-protected equipment and wiring systems (UEEHA0025).
- Detailed, visual and close inspection of hazardous-area installations (UEEHA0020, UEEHA0038).
- Maintenance of Ex equipment and periodic inspection and maintenance programs (UEEHA0026, UEEHA0023, UEECD0011).

Low Voltage Design Fundamentals and Caneco BT Training - Lokelect Engineering

September 2026 | 40 hours | Online | Certificate ID 2b3376a9aea18854

- LV design: earthing systems, power balance, cable sizing, voltage drop, selectivity basics.
- Caneco BT: project setup, modeling, results analysis and design iteration.

PROFESSIONAL EXPERIENCE

Electrical Engineer - RMBTECH EURL

March 2025 - Present | Algeria

- SCADA gas pipeline projects (Sonatrach): led electrical engineering, including a site audit of 66 isolated solar/grid-powered stations on the GK3 gas pipeline, deployment on GK1/GK2, and pre-engineering of GR1/2/4/6 (integrator: SUPCON).
- Critical equipment and ATEX: engineering for the replacement of an 8500 A rectifier for the AOA electrolysis system, and design of an ATEX Ex "p" pressurized cabinet for Sonatrach GP2Z (Arzew).
- Industrial rehabilitation: technical direction, electrical redesign (EPLAN), rewiring and automated commissioning (Siemens S7-1200) of a BOSSAR packaging machine for GBFoods (Jumbo).
- Agro-industry: electrical design for the SCADA system of the OAIC grain silo in Mesra.

Lead Electrical Engineer - PROSKID EURL

July 2023 - March 2025 | Algeria

- Designed and sized electrical systems of Oil & Gas skids for clients ENGCB/GCB and SARPI.
- Produced power supply, power and control schematics in EPLAN Electric P8 and AutoCAD Electrical.
- Designed and sized the off-grid solar power supply systems feeding the skids.
- Commissioned 27 Milton Roy chemical anticorrosion dosing skids and their solar installations.
- Designed and built an ATEX Ex "d" flameproof enclosure for a corrosion inhibitor skid, including component mounting and wiring.

Photovoltaic Design Engineer - Sungy SPA

October 2022 - July 2023 | Algeria

- Delivered over 300 kWp of installed solar capacity: 3D layout, sizing, simulation, electrical diagrams.
- Commissioning and monitoring of solar installations; energy sobriety plans; deployable mobile solar solutions.

Document Controller - CEVIELEC SPA (Cevital Electrique)

March 2022 - October 2022 | Algeria

- ALRAR Gas Processing Complex re-instrumentation project.
- Cabinet wiring, instrument installation, loop checks and troubleshooting.
- Management of revisions, modifications and document markups; project planning and tracking.

FREELANCE

5.4 MW Industrial Solar Installation - SARL Ibrahim & Fils (IFRI), Akbou, Bejaia

- Rooftop PV design for a zero-injection self-consumption plant: Huawei SUN2000-150K inverters, 10 x 241 kWh storage; single-line diagrams, load and power quality analysis, DC/AC cable sizing.

Electrical Training Simulations - ROAR Training Solutions (Australian RTO)

- 3D playable electrical competency exercises (substation switching, Low Voltage Rescue and CPR) in Blender and Godot 4, delivered as SCORM 1.2 browser modules.

EDUCATION

State Engineer + Master II, Electrical Engineering

Ecole Nationale Polytechnique d'Oran - Maurice Audin (ENPO-MA), Oran, Algeria | 2016 - 2021

- Valedictorian, class of 2018-2021.
- Master's thesis: design, build and commissioning of a 5 kW autonomous PV microgrid (3 kW PV, 800 Ah battery bank, 50 kVA genset, three-source hybrid inverter).

LANGUAGES

Arabic - Native | French - Bilingual, C1 certified (DALF 76/100) | English - Fluent